

Pratyush Talwar

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TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript

Frontend: HTML, React.js, Tailwind CSS

Backend/Databases: Node.js, Express.js, MongoDB, MySQL

DevOps & Cloud: AWS EC2, Linux, Ansible, Git, Docker, YAML, Postman

Core Subjects: Data Structures and Algorithms (DSA), DBMS, OOPS, Operating Systems, Distributed Systems

PROJECTS

AI Chat Interface | *React.js, Node.js, Anthropic API, Tailwind CSS*

- Built a full-stack AI-powered chat application integrating Anthropic's Claude API with a responsive React frontend and Express.js backend.
- Implemented streaming responses, conversation history management, and session persistence using MongoDB, supporting 100+ concurrent sessions.
- Deployed on AWS EC2 with Docker containerization, achieving sub-200ms average response initiation latency and 99.9% uptime.
- Designed a clean, accessible UI with Tailwind CSS featuring dark mode, message threading, and markdown rendering for code snippets.

Real-Time Code Collaboration Platform | *React.js, Node.js, Socket.io, MongoDB*

- Engineered a real-time collaborative code editor supporting 20+ simultaneous users per room with conflict-free operational transformation (OT) for edits.
- Integrated syntax highlighting for 10+ languages using CodeMirror and built a secure JWT-based authentication system for room management.
- Implemented persistent session storage with MongoDB, allowing users to resume coding sessions and review full edit history.
- Optimized WebSocket event handling to reduce synchronization lag to under 50ms across geographically distributed users.

DevOps Monitoring Dashboard | *Python, React.js, Docker, Ansible, AWS EC2*

- Developed an infrastructure monitoring dashboard that aggregates CPU, memory, disk, and network metrics from 15+ EC2 instances in real time.
- Automated server provisioning and configuration management using Ansible playbooks, reducing deployment time by 60%.
- Containerized all microservices with Docker Compose and set up CI/CD pipelines using GitHub Actions for zero-downtime deployments.
- Built alerting workflows that trigger Slack/email notifications on threshold breaches, reducing mean time to detection (MTTD) by 40%.

EDUCATION

Delhi Technological University

Bachelor of Technology — Information Technology

Current CGPA: 7.39

New Delhi, India

2024 – Present

K.R. Mangalam World School

Higher Secondary (Class XII)

Percentage: 92.6%

New Delhi, India

Passed 2024

ACHIEVEMENTS

- JEE Mains 98.9+ Percentile** — Ranked among top 1% of 1.2 million+ candidates nationally
- 300+ LeetCode Problems Solved** — Consistent problem solver across Easy, Medium, and Hard categories
- School Head Boy** — K.R. Mangalam World School — elected to lead student body and represent school at events

- **FIDE Rated Chess Player** — Rating 1440 — competed in rated tournaments under FIDE (International Chess Federation)
- **Coding Ninjas — Top Performer** — Certificate of Excellence for DSA & Introduction to Java (October 2025)